

Automated Detection of Illegal Gold-Mining Barges (Garimpo) on the Rio Madeira Using Optical Satellite Imagery

Progress Report II — Research Internship

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Abstract

This report documents the second stage of a joint EMINES UM6P–UFAM research internship building an automated, learning-based detector for garimpo (illegal gold-mining) activity on the médio Rio Madeira from CBERS 4A optical imagery, extending the manual-digitizing methodology of Monteiro (2024) and the first-stage pipeline reported previously. Since the previous report, the dataset has been re-audited and corrected (1,292 usable tiles, 38 positive / 1,254 negative, following the removal of one label found to sit on a mosaic-seam artifact), a supervised segmentation model has been designed, trained, and evaluated (a pretrained ResNet18 U-Net predicting a per-pixel garimpo probability heatmap), and a full-scene sliding-window inference pipeline has been built and validated: on the held-out validation footprint, every one of five known garimpo sites is recovered with near-maximal confidence and no false positives elsewhere in the scene, while a scan of a scene with zero labeled sites surfaces a small number of candidate false positives, most of which trace to a documented image-artifact sensitivity and a minority of which remain genuinely ambiguous. Aggregated across all 13 available scenes, the detector achieves 100% recall and 30.4% raw precision (43.6% once artifact-linked false positives are excluded) at a probability threshold of 0.5. A first prototype interactive map interface has also been built, overlaying model output on a real satellite basemap using footprint coordinates recovered from CBERS product metadata. This report details the methodology, the diagnostic process behind each of these results, and the concrete next steps required before the pipeline can be considered a reliable detection tool.

1 Introduction

Gold mining along the Rio Madeira has been an active, continuous practice since at least the early 1980s, evolving from small manual operations on the riverbanks into a mechanized industry built around suction dredges mounted on floating barges (Bastos and Lacerda, 2004). Because extraction takes place underwater, this activity leaves almost none of the surface signatures typically associated with terrestrial mining (e.g. exposed pits, tailings piles), which makes remote detection considerably harder than for land-based illegal mining. The barges and dredges themselves — along with the sediment plumes, sandbanks and bar deposits they leave behind — are in practice the only reliably observable indicators of the activity from space.

This project is carried out as part of a research internship agreement between the EMINES School of Industrial Management (UM6P), Morocco, and the Departamento de Geociências (DEGEO), Universidade Federal do Amazonas (UFAM), Brazil. It builds directly on the undergraduate thesis of Anna Beatriz de Oliveira Monteiro (Monteiro, 2024), which established the manual methodology and generated four years (2020–2023) of visually-digitized barge locations along a stretch of the médio Rio Madeira. This report covers the CBERS 4A optical detection track specifically; a parallel Sentinel-1/Sentinel-2 fusion track, developed by other members of the internship team, is reported separately.

1.1 Objectives

- Build a reproducible image-processing pipeline that isolates the river channel and produces analysis-ready tiles from raw CBERS 4A scenes.
- Construct a labeled image dataset (garimpo present / absent, with bounding boxes) from the available batch of CBERS 4A scenes, and audit it for data-quality issues before training.
- Train and evaluate a small-data-appropriate segmentation model that outputs a per-pixel garimpo probability heatmap, rather than a classification label alone.
- Extend detection from individual tiles to full, arbitrarily large scenes via sliding-window inference, and quantify precision/recall across the full available dataset.
- Build a first prototype for visualizing detector output on a real geographic map, as a step toward a deployable monitoring tool.

2 Study Area and River Characteristics

The study area is located on the médio Rio Madeira, in the municipality of Manicoré, Amazonas State, Brazil, within the same reach studied by Monteiro (2024). Table 1 summarizes the key physical characteristics of the Rio Madeira basin relevant to this work.

Table 1: Key physical characteristics of the Rio Madeira basin.

Characteristic	Value / Description
Drainage basin area	$\sim 1,324,727 \text{ km}^2$ (largest Amazon sub-basin)
Share of Amazon basin	$\sim 23\%$
Mean annual discharge	$\sim 26,500 \text{ m}^3/\text{s}$
Origin	Andes Cordillera (Beni, Madre de Dios, Mamoré rivers)
Hydrological regime	Tropical austral (single annual flood peak, $\sim$ April; low water $\sim$ September–October)
Channel pattern	Anabranching, locally straight to meandering; straight reaches up to $\sim 35 \text{ km}$
Water characteristics	pH 6.5–7, high electrical conductivity, high suspended sediment load
Gold transport mechanism	Secondary (alluvial) gold, transported as bedload/suspended sediment from Andean source rocks

The strong seasonal (vazante/enchente) cycle is particularly relevant to this project: barge activity is known to be inversely related to water level, with more barges operating during low water (roughly July–December) than during the flood peak (Monteiro, 2024).

3 Prior Work: Monteiro (2024)

The foundation for this project is the undergraduate thesis of Monteiro (2024), developed at UFAM under the supervision of Prof. Naziano Pantoja Filizola Júnior. Her work is, to our knowledge, the first systematic attempt to identify and quantify garimpo barges on the médio Rio Madeira using a combination of optical (CBERS 4A) and radar (Sentinel-1A) imagery.

3.1 Methodology summary

Monteiro processed CBERS 4A WPM imagery in QGIS (RGBN band composition followed by pansharpening with the 2 m panchromatic band), and processed Sentinel-1A dual-polarization (VV/VH) imagery in SNAP, generating three derived RGB products (Dual Pol Ratio, Dual Pol Difference, Dual Pol Multiple). Barges and dredges were then identified and digitized manually, as point vector features, directly in QGIS — no automated or learning-based detection was used at any stage.

3.2 Key findings

- CBERS 4A optical imagery, at 2 m pansharpened resolution, allowed clear visual identification and counting of barges, though it could not distinguish barges from dredges specifically.
- Sentinel-1 SAR was able to detect some metallic-hulled vessels as bright backscatter points even under cloud cover, but detected far fewer vessels overall than the optical imagery in the available image dates.
- Barge counts varied strongly by season and year, ranging from as few as 24 (June 2021) to as many as 241 (June 2020) across the study reach.
- Barge locations correlated spatially with fluvial bars and alluvial deposits, consistent with the known alluvial origin of Madeira River gold.

4 Remote Sensing Background

4.1 How satellite remote sensing informs this problem

Every satellite sensor used in this project trades off three resolutions: spatial resolution, spectral resolution, and temporal resolution. Detecting a garimpo barge — an object typically only a few meters to a few tens of meters long — is fundamentally a spatial-resolution problem: it requires imagery fine enough to resolve a small target against a highly textured water background.

4.2 Why CBERS 4A at 2 m resolution

CBERS 4A WPM pansharpened imagery (2 m spatial resolution) was selected as the primary data source for this project for three reasons: (1) resolution adequacy — a barge on the order of 10–20 m in length spans roughly 5–10 pixels; (2) continuity with prior work — using the same sensor as Monteiro (2024) allows direct comparison on overlapping dates; (3) cost and access — CBERS 4A imagery is freely distributed by INPE.

5 Data and Methodology

Figure 1 gives an overview of the full processing pipeline described in this section and Sections 7–9, from raw scene acquisition through to the interactive map prototype.

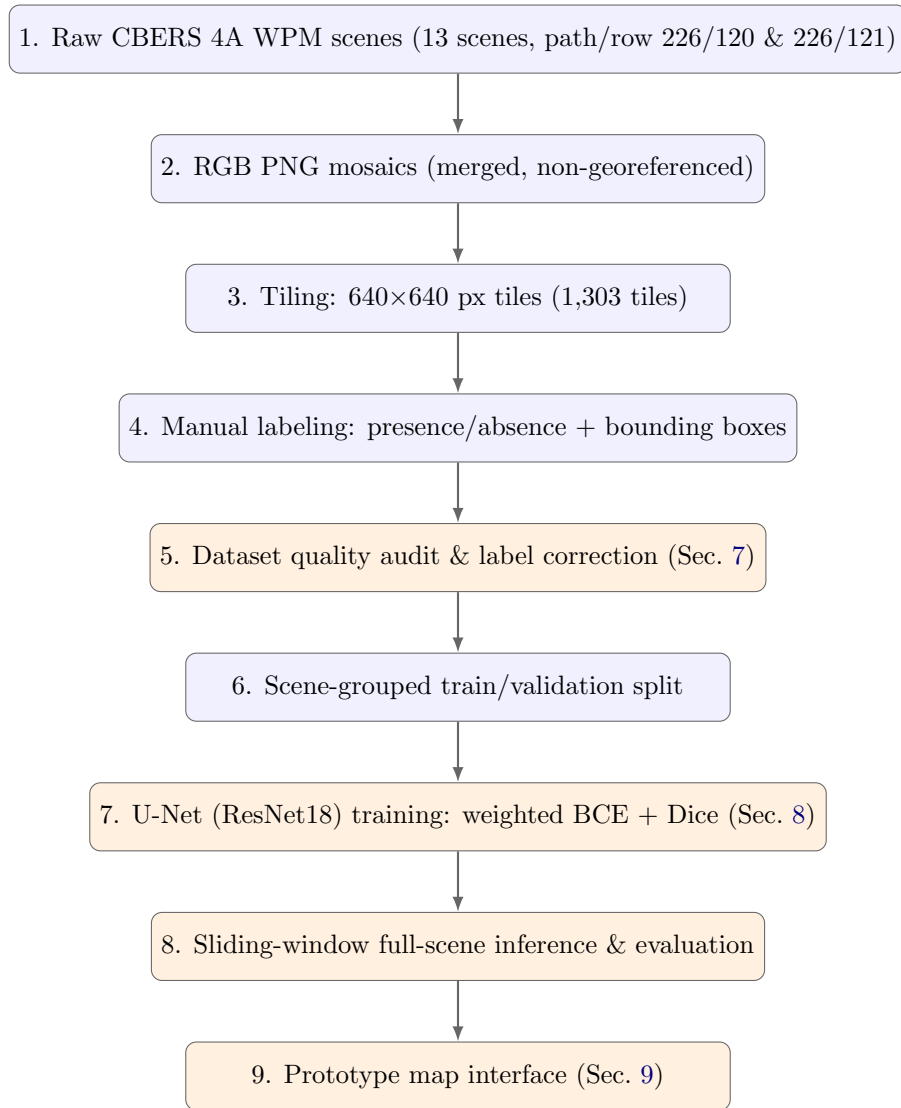


Figure 1: Overview of the full processing pipeline. Orange stages (5, 7–9) are new since the previous report.

5.1 Data acquisition and format

Twelve CBERS 4A WPM scenes were originally acquired covering path/row 226/120 and 226/121 across multiple acquisition dates between 2020 and 2023. A subsequent, more detailed count of the raw scenes revealed a thirteenth scene, 2021-10-15 (226/121), which contains zero labeled positive tiles and had therefore not been separately counted in the earlier scene inventory; it is included in the present analysis as a genuine true-negative scene and proves useful as a specific diagnostic case later in this report (Section 8.4).

During pipeline development, the original raw 4-band GeoTIFFs became unavailable locally (only RGB-merged, non-georeferenced PNG mosaics remained from an earlier processing pass). A deliberate scope decision was made to proceed with model development on this RGB PNG data — without the near-infrared band and without native georeferencing — rather than lose development time re-acquiring the raw imagery. This is documented explicitly because it shapes several downstream design choices (Section 8) and is revisited as a concrete next step (Section 10). The original georeferenced 4-band TIFF products (BAND0–BAND4, with accompanying XML metadata) were subsequently located in a project data mirror and used later in this report specifically to recover approximate scene footprint coordinates for the prototype map interface (Section 9); they have not yet been reintroduced into the model training pipeline itself. Figure 2 shows a representative full raw scene.

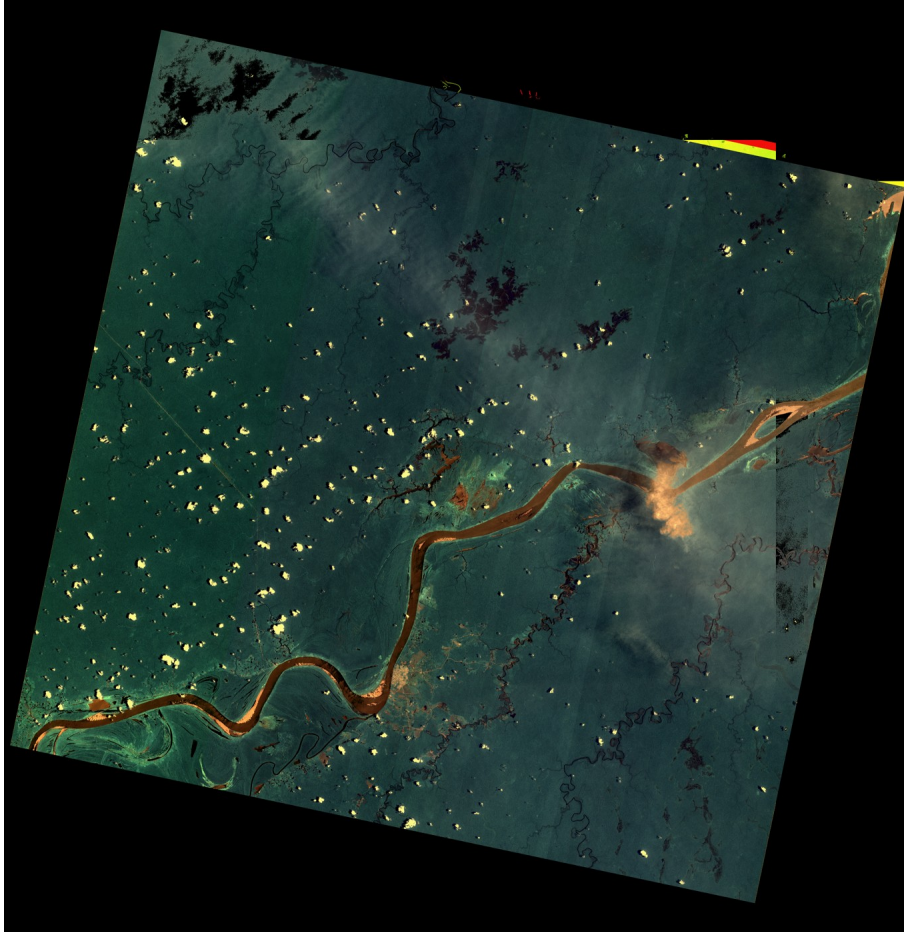


Figure 2: Representative full CBERS 4A WPM scene (15 October 2021, path/row 226/120), as merged to RGB PNG. The rotated diamond-shaped content region against black padding reflects the sensor’s actual (non north-aligned) scan geometry within a north-up padded export canvas, confirmed against CBERS product metadata in Section 9.

5.2 Tiling

Each of the merged RGB PNG mosaics was divided into fixed 640×640 pixel tiles. This produced 1,303 tiles in total across the 13 scenes.

5.3 Labeling scheme

Tiles are labeled with a three-way scheme: *positive* (at least one garimpo barge/dredge visually identifiable), *negative* (no barge visible), or *uncertain* (pending review, excluded from training and evaluation). Positive tiles are additionally annotated with pixel-space bounding boxes around each visible barge/dredge.

6 Labeling Results

Manual labeling was completed for all 1,303 tiles. Table 2 gives the final counts, following a data-quality audit and one label correction detailed in Section 7.

Table 2: Final labeling results, post-audit.

Category	Count
Tiles generated	1,303
Uncertain (excluded from training/evaluation)	11
Usable tiles (positive + negative)	1,292
Negative	1,254
Positive	38
Bounding boxes drawn (positive tiles)	42

The overall positive rate is low ($\sim 2.9\%$ of usable tiles), consistent with the sparse, mobile nature of garimpo barges relative to the total river-corridor area covered. At the pixel level the imbalance is far more extreme: across the training split alone, positive pixels make up approximately 0.0025% of all pixels (a raw negative:positive pixel ratio of roughly 40,000:1), which directly shapes the loss design in Section 8.

7 Dataset Quality Audit

Before committing to a training run, the labeled dataset was audited for a specific, plausible failure mode: garimpo boxes located near mosaic seams or sensor nodata regions, which could let a model learn to associate the *artifact* rather than the actual garimpo signature with a positive label.

7.1 Methodology

For every positive-tile bounding box, the nearest sufficiently large (>500 px) near-black connected region in the same tile was located, and the pixel distance from the box to that region was computed. Boxes within 100 px of such a region were flagged for manual visual review. Flagged blobs were further split into two categories: those touching the tile’s outer border (consistent with a genuine mosaic/nodata seam) and those fully interior to the tile (more consistent with an ordinary cloud shadow, which is common, benign image content rather than a data defect).

7.2 Findings

An initial pass flagged 10 of 43 boxes (across 39 positive tiles) within the 100 px threshold; 7 of these touched a tile border and 3 were interior. Manual visual inspection of all 7 border-touching cases confirmed a real mosaic-seam or no-data artifact adjacent to the labeled box in each case, with one case in particular — a box only 1.41 px from the seam boundary, visibly on the dark side of a hard stitching line — judged unambiguous enough to warrant exclusion. The remaining 6 border-touching cases, and the 3 interior (cloud-shadow-adjacent) cases, were kept: a plausible, if not provable, explanation exists for each (garimpo sites plausibly cluster near river-adjacent scene boundaries for genuine geographic reasons; cloud shadows are ordinary scene content), and removing labels on a hunch would trade one bias for another.

The one clearly contaminated label (tile `.._20230728_226_120_.._x512.y4096.png`, box coordinates (270, 477)–(292, 512)) was reclassified from positive to negative, its box removed, and its rasterized mask regenerated as all-zero. A repeat of the audit on the corrected dataset confirms the fix: 9 of 42 remaining boxes flagged, 6 border-touching, 3 interior — exactly one fewer than before, and no other counts perturbed. This finding, and the full per-box distance table, is documented in `docs/dataset_limitations.md` in the project repository for methodological transparency.

8 Deep Learning Detector

8.1 Architecture and loss design

A U-Net-style segmentation model was chosen over a bounding-box object detector, since the pixel-space bounding-box labels can be rasterized directly into training masks (filled rectangles) and the resulting model output — a per-pixel probability map — matches the desired final deliverable (a continuous garimpo-probability heatmap) without any post-processing conversion step.

The encoder is a ResNet18 backbone pretrained on ImageNet, fully fine-tuned (not frozen), with a 3-channel RGB input (matching the available data; Section 10 discusses reintroducing the near-infrared band). A pretrained encoder was preferred over training from scratch given the limited data budget (1,002 training tiles, 30 positive): low- and mid-level convolutional filters learned on natural images transfer usefully even under the domain shift to satellite RGB imagery, and a from-scratch encoder would spend much of its limited data budget relearning generic filters. ResNet18 specifically (rather than a deeper ResNet34/50) was chosen to limit overfitting risk given how few positive examples are available.

The loss function combines weighted binary cross-entropy and Dice loss in equal proportion. Given the extreme pixel-level class imbalance (Section 7), a positive-class weight for the BCE term is computed directly from training-set pixel statistics each run, then capped at 50 to prevent gradient instability from the raw $\sim 40,000:1$ ratio; Dice loss, being scale-invariant to class frequency, does most of the imbalance-handling work. All three loss components (BCE, Dice, and the combined objective) are logged separately per epoch for diagnosis.

Formally, let z_i denote the model’s raw output (logit) at pixel i , $p_i = \sigma(z_i) = 1/(1 + e^{-z_i})$ the predicted probability after the sigmoid, and $y_i \in \{0, 1\}$ the ground-truth label at that pixel, over the N pixels of a tile. The weighted binary cross-entropy is

$$\text{BCE} = -\frac{1}{N} \sum_{i=1}^N \left[w_{\text{pos}} y_i \log(p_i) + (1 - y_i) \log(1 - p_i) \right], \quad (1)$$

which is simply the average, per-pixel penalty for how wrong and how confident each individual prediction was, with positive-class pixels counted w_{pos} times more heavily than negative ones. The weight itself is computed from the training split each run rather than fixed by hand,

$$w_{\text{pos}} = \min\left(\frac{N_{\text{neg}}}{N_{\text{pos}}}, 50\right), \quad (2)$$

where N_{neg} and N_{pos} are the total negative and positive pixel counts across the training masks; the uncapped ratio is approximately 40,000:1 (Section 7), so the cap at 50 is strongly binding in practice — without it, the rare positive pixels would dominate the gradient by a factor large enough to destabilize training.

The Dice coefficient, by contrast, does not treat pixels independently; it measures the overlap between the predicted and true positive regions as a whole,

$$\text{Dice} = \frac{2 \sum_{i=1}^N p_i y_i + \epsilon}{\sum_{i=1}^N p_i + \sum_{i=1}^N y_i + \epsilon}, \quad \mathcal{L}_{\text{Dice}} = 1 - \text{Dice}, \quad (3)$$

with a small smoothing constant ϵ added to both numerator and denominator to keep the loss well-defined on all-negative tiles (where $\sum y_i = 0$). Unlike BCE, Dice is computed entirely from the positive pixels and their overlap — the vast negative background contributes nothing to it once the model is confidently predicting near-zero there, so it does not benefit from the same easy, aggregate improvement that drives BCE down quickly in early training (Section 8, and Figure 4 below). The two are combined in equal proportion,

$$\mathcal{L} = \lambda_{\text{BCE}} \cdot \text{BCE} + \lambda_{\text{Dice}} \cdot \mathcal{L}_{\text{Dice}}, \quad \lambda_{\text{BCE}} = \lambda_{\text{Dice}} = 0.5, \quad (4)$$

and it is exactly this combination — an average over N pixels dominated numerically by easy negatives (Eq. 1) alongside a term that only ever looks at the sparse positive region (Eq. 3) — that produces the checkpoint-selection behavior documented in Section 8: the two terms are not measuring the same thing, and do not necessarily reach their best value at the same training epoch.

Conservative, geospatially-appropriate data augmentation is applied during training only: horizontal/vertical flips, 90° rotations, and small brightness/contrast jitter, with spatial transforms applied jointly to image and mask and photometric transforms applied to the image only. A weighted sampler oversamples positive tiles during training, since plain random shuffling at batch size 8 would yield an expected 0.25 positive tiles per batch given the base class distribution.

8.2 Training infrastructure and data split

Initial CPU benchmarking (single forward/backward pass, 640×640 input) measured approximately 1 second per sample, projecting to roughly 6–8 hours for a full 20-epoch training run — impractical for iterative development. Training was therefore moved to a GPU-backed environment (Google Colab, free tier), reducing full 20-epoch runs to approximately 40–45 minutes each.

The dataset was split at the *scene* level, not the tile level, to avoid leakage between spatially or temporally adjacent tiles. Given only 38 positive tiles spread across 13 scenes, an exhaustive search over all scene subsets (computationally trivial at this scale) was used to find the train/validation partition minimizing combined deviation from a target 80/20 split, on both tile-count and positive-count fraction, while rejecting any candidate with zero positive tiles in validation. This produced a 10-scene training split (1,002 tiles, 30 positive) and a 3-scene validation split (290 tiles, 8 positive).

A specific limitation was identified and documented: the dataset spans only two distinct geographic footprints (path/row 226/120 and 226/121), each reimaged at roughly six different dates. No scene-level split — however well balanced — can therefore be fully independent in the geographic sense; the same river bends and mining sites can appear in both the training and validation splits at different dates. A fully footprint-independent split exists (all of one footprint in validation) but would push the validation fraction to 38.5% of all tiles, an unacceptable

reduction in training data given how few positive examples exist. Given this constraint, and given that a genuinely independent held-out test set would inherit the same limitation, a three-way train/validation/test split was deliberately not pursued; validation performance is reported as a development signal useful for model selection and gross-overfitting detection, understood to be a measure of generalization across imaging *dates* at fixed locations rather than to genuinely new locations.

8.3 Training results and checkpoint selection

Three full training runs were performed (20 epochs each, fixed seed), the first two using the pre-audit labels and the third using the corrected labels from Section 7. Figure 3 shows the final run’s training curves.

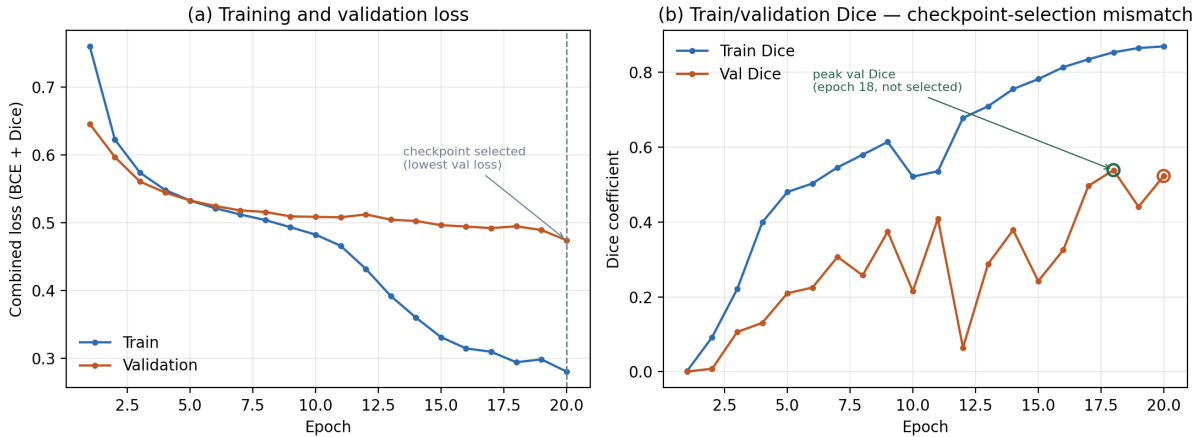


Figure 3: Training curves for the final run (corrected labels, 20 epochs). (a) Combined loss decreases smoothly and near-monotonically on both splits; the checkpoint with the lowest validation loss (epoch 20) is selected as final. (b) Dice coefficient on the same run: training Dice increases steadily to 0.87, while validation Dice is markedly noisier and peaks earlier, at epoch 18 (0.539), one epoch before the checkpoint actually selected (epoch 20, 0.523).

A specific and repeatable methodological finding emerged across all three runs: the checkpoint selected by lowest validation combined loss is consistently *not* the checkpoint with the highest validation Dice score, and the gap between them varies run to run (peak validation Dice occurred at epoch 10, 15, and 18 in the three respective runs, never at the final, selected epoch). This is explained by the validation set’s small size — only 8 positive tiles — which makes Dice, a metric computed only over the sparse positive-pixel region, highly sensitive to small changes in a handful of predictions, whereas the combined loss is dominated by the binary cross-entropy term, an aggregate over all pixels (99.9975% of which are negative) and therefore far smoother epoch to epoch. Checkpoint selection on validation combined loss, rather than raw Dice, is judged the more defensible choice specifically because of this instability, though it is worth noting explicitly that the two criteria do not always agree on which model is best, and that a model selected this way is not guaranteed to be the one with maximal segmentation overlap quality on any given run.

Figure 4 isolates the two loss components to make the mechanism behind this finding directly visible: BCE (Eq. 1) decreases smoothly and almost monotonically from the first epoch — since it is an average over N pixels, and the model can lower that average substantially just by becoming confidently correct on the overwhelming negative majority, well before it has learned anything useful about the rare positive class — while Dice loss (Eq. 3) remains close to its maximum (worst) value for roughly the first ten epochs before beginning to improve, since it is computed only from the positive-pixel overlap and gives the model no credit at all for correct

negative predictions. Since the combined loss (Eq. 4) is an equal-weighted average of the two, its early- and mid-training trajectory is effectively governed by BCE alone, which explains why checkpoints selected by combined loss can diverge from the checkpoint with the best actual segmentation overlap.

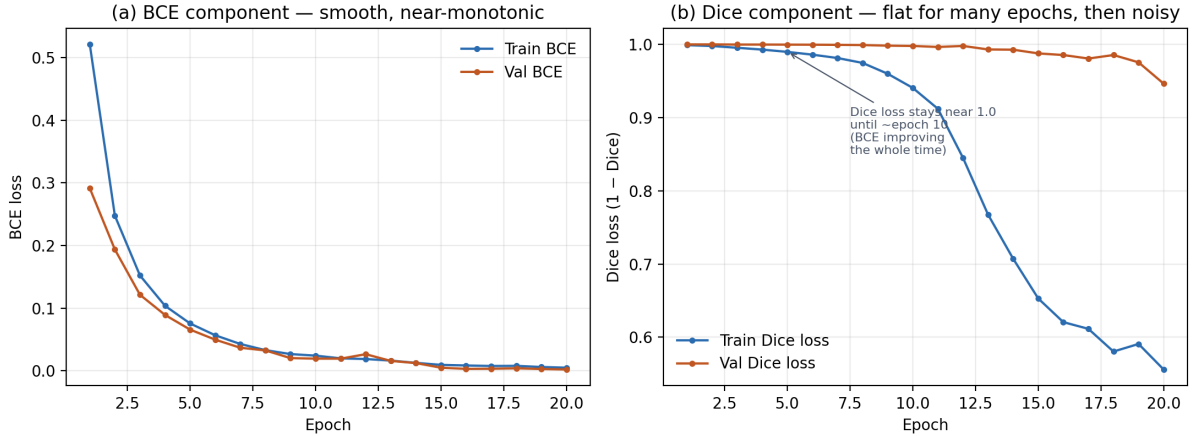


Figure 4: Decomposition of the combined loss into its BCE and Dice components (final run, corrected labels). The BCE term drives early optimization; the Dice term only becomes informative once the model has already learned to concentrate probability mass on the rare positive class.

This checkpoint-selection mismatch was not a one-off observation: it recurred, in varying degree, across all three full training runs performed during this project (the first two using pre-audit labels, the third using the corrected dataset from Section 7), summarized in Table 3.

Table 3: Comparison across all three full training runs. “Selected” refers to the checkpoint chosen by lowest validation combined loss (the deployed model in each run); “Peak Dice” refers to the epoch with the highest validation Dice score observed anywhere in that run.

Run	Labels	Selected epoch	Selected val Dice	Peak Dice epoch	Peak val Dice
Run 1	pre-audit	20	0.461	10	0.556
Run 2	pre-audit	20	0.570	15	0.647
Run 3	post-audit	20	0.523	18	0.539

In every run, the selected checkpoint is the final epoch (20), while the peak-Dice epoch varies unpredictably run to run (10, 15, then 18) — a direct consequence of validation Dice being computed over only 8 positive tiles, and further evidence that raw Dice, on a validation set this small, should be treated as a noisy diagnostic signal rather than a reliable epoch-selection criterion. Notably, the gap between the selected and peak Dice scores narrows substantially in Run 3 (relative gap $\sim 3\%$, versus $\sim 19\%$ and $\sim 14\%$ in Runs 1 and 2); this is consistent with, though not proven to be caused by, the single label correction applied between Run 2 and Run 3, given the single-run sample size.

8.4 Full-scene inference and evaluation

A sliding-window inference pipeline was built to apply the tile-trained model to full, arbitrarily large scene images: overlapping 640×640 windows are slid across the scene, the model is run on each window independently, and overlapping predictions are averaged and stitched into one continuous probability heatmap the size of the original scene.

On the held-out validation scene 20220921.226_120 (five known garimpo sites), every one of the five sites was recovered with maximum predicted probability (1.00) inside its ground-truth box, and a scan of the entire scene for probability-0.5 blobs found exactly five, matching the five known sites with no additional false positives anywhere else in the scene. Figure 5 shows one such example: the predicted heatmap is visibly faint at the scale of a full 640×640 tile, but a tight crop around the ground-truth box shows a clear, correctly-located confidence response, consistent with the quantitative result.

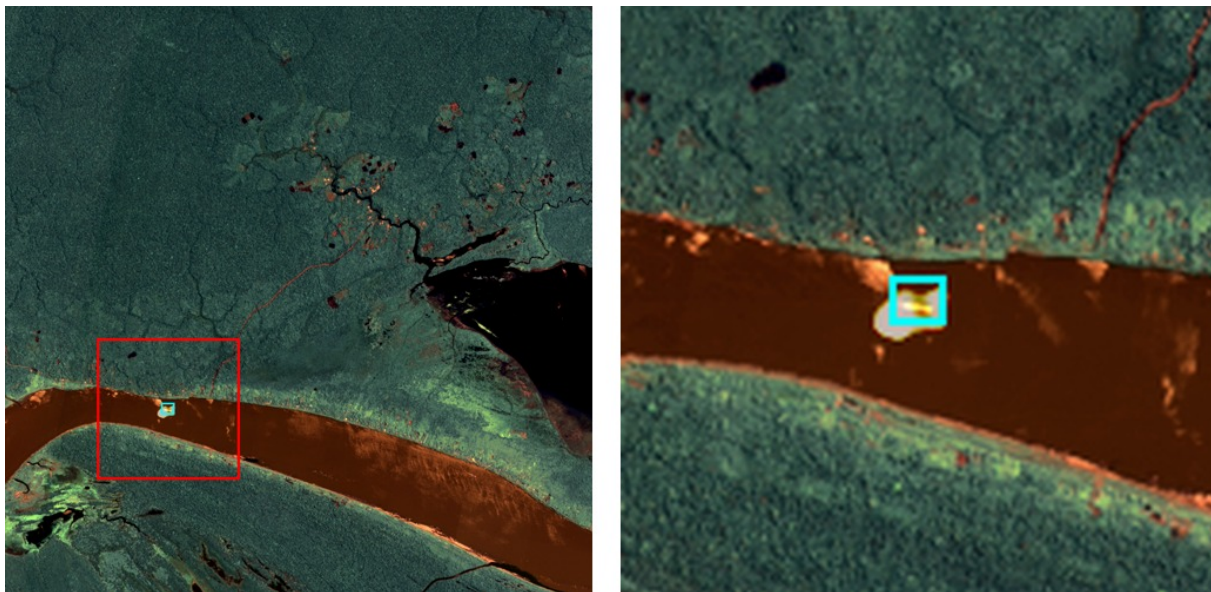


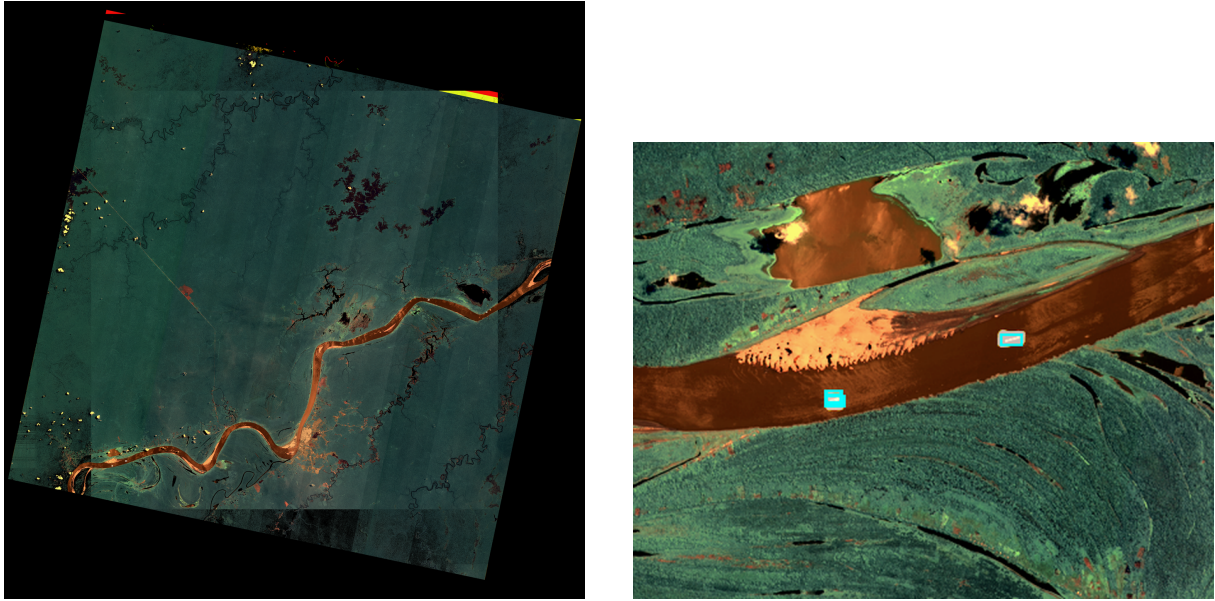
Figure 5: Example detection result on a held-out validation tile (20220921.226_120). Left: full tile, with the ground-truth box (cyan) and the cropped region (red) indicated. Right: 4× zoom on the crop, showing the predicted probability heatmap (pale/orange glow) correctly centered on the ground-truth box.

On the scene with zero labeled positives, 20211015.226_121, the same scan found seven candidate blobs. Four were substantially coincident with nodata/scene-border pixels, consistent with the artifact sensitivity documented in Section 7. The remaining three were visually similar in character to confirmed detections (a bright spot at a small tributary stream-bend) but sit on data with no ground-truth label at all, so their status cannot be resolved from the available data alone. A cross-date check — scanning the same pixel coordinates across the other five acquisition dates of the same footprint — found activation only on the source date, in every case. This rules out one plausible explanation (a static, terrain-shape-based shortcut, which would be expected to fire identically on every date of the same, unchanged terrain) but does not confirm the alternative (a genuine, undocumented mining site); it is reported here as an open, unresolved finding rather than a settled one.

Figure 6 shows the full stitched heatmap for the validation scene discussed above, alongside a zoomed inset on its main detection cluster (four of the six known boxes in this scene sit within approximately 150 px of one another, near a tiling-grid corner). The zoomed view also makes visible, incidentally, the duplicate-labeling artifact identified in Section 7: overlapping ground-truth box outlines at a shared tile boundary, corresponding to the same physical site independently boxed in two adjacent tiles before the box-merging correction described there.

Applying the same evaluation across all 13 available scenes, and after correcting for a small number of duplicate labels (adjacent tiles sharing a grid boundary independently boxing the same physical site, merged via literal-overlap detection: 42 raw boxes reduced to 34 unique sites), yields the precision/recall summary in Table 4 and Figure 7.

Recall is exactly 100% across every split, a result confirmed to follow directly from the earlier



(a) Full stitched probability heatmap over the entire scene.

(b) Zoomed inset on the main detection cluster, with ground-truth box outlines (cyan) overlaid.

Figure 6: Full-scene sliding-window inference result, validation scene 20220921_226_120. The isolated colored strip near the top edge of (a) is a data-quality flag baked into the source imagery itself (confirmed by direct pixel inspection, not a model prediction).

Table 4: Detection performance across all 13 scenes, threshold = 0.5.

Split	TP	FP	FN	Precision	Recall
All 13 scenes	34	78	0	0.304	1.000
Training scenes	28	66	0	0.298	1.000
Validation scenes	6	12	0	0.333	1.000

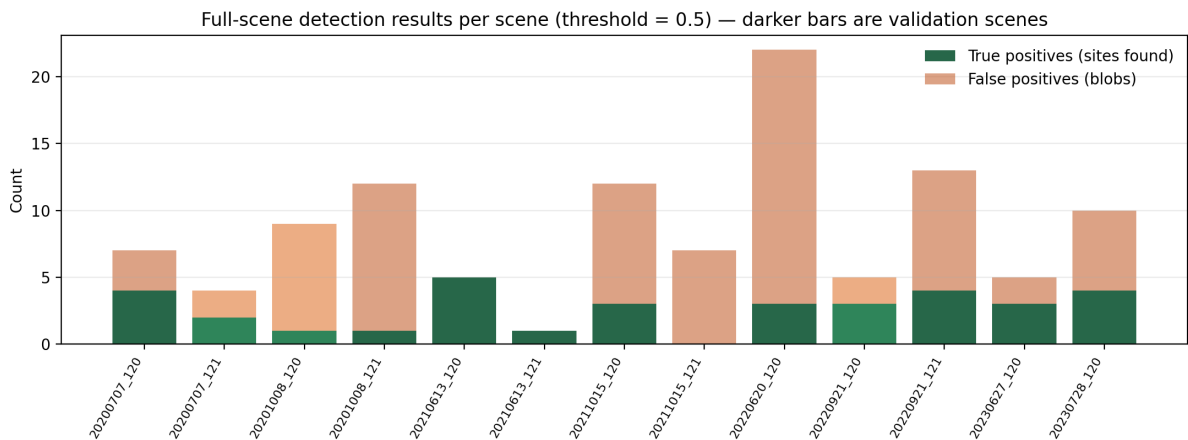


Figure 7: True and false positives per scene at threshold 0.5. Darker bars indicate validation-split scenes. The single largest source of false positives, scene 20220620_226_120 (19 of 22 detections false), is an outlier not yet explained and flagged for follow-up.

per-tile checks (every known site had already been independently confirmed to overlap a detected blob before scene-level merging), rather than an artifact of a lenient matching threshold. Of the 78 total false positives, 34 coincide with the documented nodata/artifact sensitivity; excluding these gives an adjusted precision of 43.6% on otherwise clean imagery. Table 5 gives the full per-scene breakdown, computed directly within the same matching pass as the TP/FP counts themselves (not derived by combining separate analysis runs), confirming the nodata/clean split exactly ($34 + 44 = 78$).

Table 5: False-positive breakdown by scene and cause, threshold = 0.5.

Scene	Split	TP	FP	FP (nodata)	FP (clean)
20200707.226_120	train	4	3	0	3
20200707.226_121	val	2	2	1	1
20201008.226_120	val	1	8	5	3
20201008.226_121	train	1	11	7	4
20210613.226_120	train	5	0	0	0
20210613.226_121	train	1	0	0	0
20211015.226_120	train	3	9	4	5
20211015.226_121	train	0	7	3	4
20220620.226_120	train	3	19	7	12
20220921.226_120	val	3	2	0	2
20220921.226_121	train	4	9	7	2
20230627.226_120	train	3	2	0	2
20230728.226_120	train	4	6	0	6

Two scenes stand out on inspection. The 20220620.226_120 outlier flagged above is confirmed to be dominated by clean (non-nodata) false positives (12 of 19), meaning its high false-positive count is not simply an extension of the already-understood artifact issue and remains an open item. Conversely, 20201008.226_121 and 20220921.226_121 both show nodata false positives outweighing clean ones (7 of 11 and 7 of 9, respectively), consistent with the documented sensitivity and not, on their own, cause for additional concern.

A closer investigation of the “clean” (non-nodata) false positives was prompted by an observation made while building the map interface (Section 9): several detections appeared, at first glance, to sit well inland with no visible water nearby, raising the concern that the model might be responding to bare or cleared land in general rather than any water-related signature. This was investigated in two stages, and the first was explicitly discarded once shown to be flawed: an initial proximity check — distance from each false positive to the nearest water pixel, using a color-based river mask built directly from the imagery and validated visually against the true river channel and its tributaries — showed roughly 41% of the 78 false positives falling within a loose 200 px buffer of water. This number is not diagnostic on its own, however: the same check applied to the 34 true positives shows them at 100% within a much tighter 30 px buffer (median distance 0 px) — essentially every detection, correct or not, falls somewhere near the river, simply because that is where both real mining activity and the model’s attention concentrate. Raw proximity cannot separate a genuine water-related confusion from ordinary river-corridor geography.

A stricter, spectral test resolves this: rather than asking whether a false positive is merely *near* water, it asks whether the false positive’s own pixels fall *inside* the calibrated water/sediment color signature. This reclassifies the false positives into mutually exclusive, fully reconciled categories, summarized in Table 6. Of the 44 clean false positives, 18 (23.1% of all 78) sit directly on water- or sediment-colored pixels — consistent with bare sediment and mining tailings sharing the same warm, low-saturation tone as exposed riverbank sediment in this sensor’s true-color

rendering, a plausible and specific causal mechanism rather than mere proximity. A further 10 (12.8%) sit adjacent to a nodata/seam region that falls just outside the detection’s own bounding box (missed by the tighter check in Section 7, which only examined the box interior), and 4 (5.1%) sit beside an interior, non-border-touching dark patch consistent with cloud shadow — a previously undocumented false-positive mechanism. The remaining 12 (15.4%) have no confident explanation under any of these checks and are reported as genuinely unresolved, rather than forced into one of the other categories. Manual review of the single highest-confidence unresolved case found it sitting at a tight bend of a narrow, dark tributary creek — plausibly still water-related, but with a color signature (dark, low-saturation) that falls outside the range calibrated on the wide, sediment-tinted main channel; the water mask’s known under-sensitivity to narrow, shaded tributaries makes the true unresolved count more likely an upper bound than a precise estimate of a distinct failure mode.

Table 6: Root-cause breakdown of all 78 false positives, threshold = 0.5. Categories are mutually exclusive and reconcile exactly to the total.

Cause	Count	% of 78
Nodata/seam within own bounding box (Section 7)	34	43.6%
Water/sediment-colored pixels	18	23.1%
Nodata/seam just outside bounding box	10	12.8%
Interior cloud shadow	4	5.1%
Unexplained	12	15.4%

Taken together, 84.6% of all false positives (66 of 78) now have a specific, visually-confirmed or spectrally-verified explanation, spanning three related but distinct mechanisms — scan/mosaic artifacts, genuine spectral similarity between mining-related sediment and natural riverbank sediment, and cloud shadow — none of which point to the model responding to arbitrary or unrelated terrain.

8.4.1 Geomorphological interpretation

The dominant false-positive mechanism identified above — spectral confusion with natural water/sediment features — is not an arbitrary model weakness; it follows directly from the fluvial geomorphology of the study reach, and connects this project’s results back to the founding observation of Monteiro (2024).

Madeira River gold is secondary (alluvial) gold, eroded from Andean source rocks and transported downstream as bedload and suspended sediment (Table 1). In a meandering, anabranching channel, flow velocity drops sharply on the inside of a bend, and it is precisely there — on the resulting *point bar* — that this sediment, including its denser gold fraction, preferentially settles. This is a well-established physical sorting process, not a mining-specific phenomenon. Figure 8 illustrates the mechanism.

This single mechanism explains three separate observations at once. First, it explains why Monteiro (2024) found manually-digitized barge locations correlating with fluvial bars: miners target point bars because that is physically where the gold concentrates, not by coincidence. Second, it explains why this project’s true positives sit essentially on the water/sediment mask (median distance 0 px, Section 8.4): the model has correctly learned to associate garimpo with this depositional signature. Third, and most usefully, it explains the model’s principal failure mode: an actively-mined point bar and an entirely undisturbed one are composed of the same pale, low-vegetation sediment, and are often close to spectrally indistinguishable in true-color RGB imagery alone. The model has therefore learned a genuine and geologically correct cue — sediment deposition zones — without yet being able to separate *active* disturbance from a merely *favorable* location. This directly motivates the near-infrared reintroduction proposed

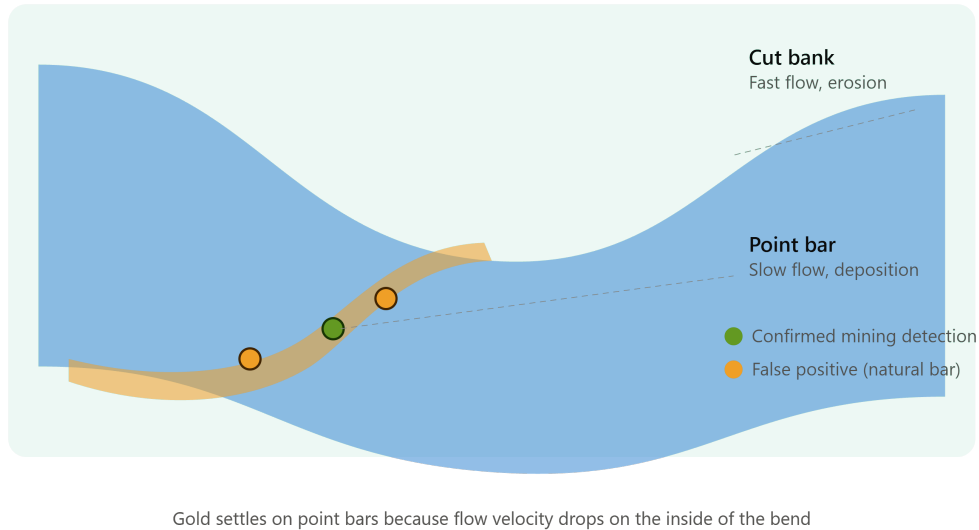


Figure 8: Schematic river meander. Flow slows on the inside of the bend, depositing sediment (the point bar, orange) — including its alluvial gold fraction — while the outer bank (cut bank) erodes under faster flow. Both real mining detections and the model’s dominant false-positive mechanism cluster on the point bar, since actively-mined and naturally undisturbed sediment share a similar true-color spectral signature.

in Section 10: disturbed, freshly-worked sediment differs from an undisturbed bar in moisture content and grain sorting in ways that are known to be more separable in the near-infrared than in visible light, offering a physically-grounded route to reducing this specific failure mode rather than a purely data-driven one.

Separately, a visually striking example encountered during interface development — a high-confidence marker appearing to sit on an unrelated inland forest clearing, well away from any labeled site — was traced to a different cause entirely: the scene in question is not one of the two dates for which exact footprint coordinates are available, and inherits the documented approximate-georeferencing limitation of Section 9 (marker *display* position offset, not a detection or model issue). Verification on an actual reference-date scene confirmed the underlying pixel-to-coordinate transform itself is correct, with only a small residual offset attributable to approximating the true rotated scene footprint with a north-aligned rectangle.

At this operating point, the detector’s error profile — never missing a known site, at the cost of a meaningful false-positive rate, now attributable in large part to specific and plausible visual/spectral confusions rather than arbitrary noise — is a reasonable fit for a first-pass screening tool, where a missed site is a materially worse outcome than an extra location flagged for human review, though the validation-split numbers in Table 4 are based on only three scenes and should not be over-interpreted as a tightly estimated figure.

9 Prototype Map Interface

A first interactive map prototype was built to display model output overlaid on a real satellite basemap, as a step toward a deployable monitoring tool. Model probability maps are converted to transparent color overlays (a black–red–yellow ramp, with opacity scaling with predicted probability) and placed on an Esri World Imagery basemap using Leaflet.

Real-world footprint coordinates were recovered from the CBERS 4A product metadata

(XML files accompanying the original band GeoTIFFs), specifically the **boundingBox** corner coordinates, which describe a near north-aligned rectangle containing the sensor’s actual (rotated) scanned swath. Precise per-date coordinates are not yet available for every scene; the current prototype reuses one date’s footprint coordinates for all other dates of the same path/row, a reasonable approximation given CBERS’s documented ± 5 km track stability, but not pixel-exact across dates. This limitation, along with the underlying data, is documented in the project repository. A date/scene selector allows switching between all 13 available scenes, each correctly re-centering the map on its footprint.

The interface was subsequently extended to surface individual detections directly, rather than the continuous heatmap alone: detected blobs (reusing the same blob-detection logic used for full-scene evaluation, Section 8.4) are rendered as clickable markers, color-coded by confidence (max probability within the blob: red > 0.8 , orange $0.5\text{--}0.8$, blue < 0.5), with a live threshold slider controlling which detections are shown. Since running blob detection directly in-browser is not practical at full scene resolution, marker positions are precomputed across a grid of 19 threshold levels ($0.05\text{--}0.95$) and the slider snaps to the nearest precomputed level, giving the appearance of continuous control in practice. Figure 9 shows the interface on a reference-date scene (2020-10-08, path/row 226/121), for which exact footprint coordinates are available.

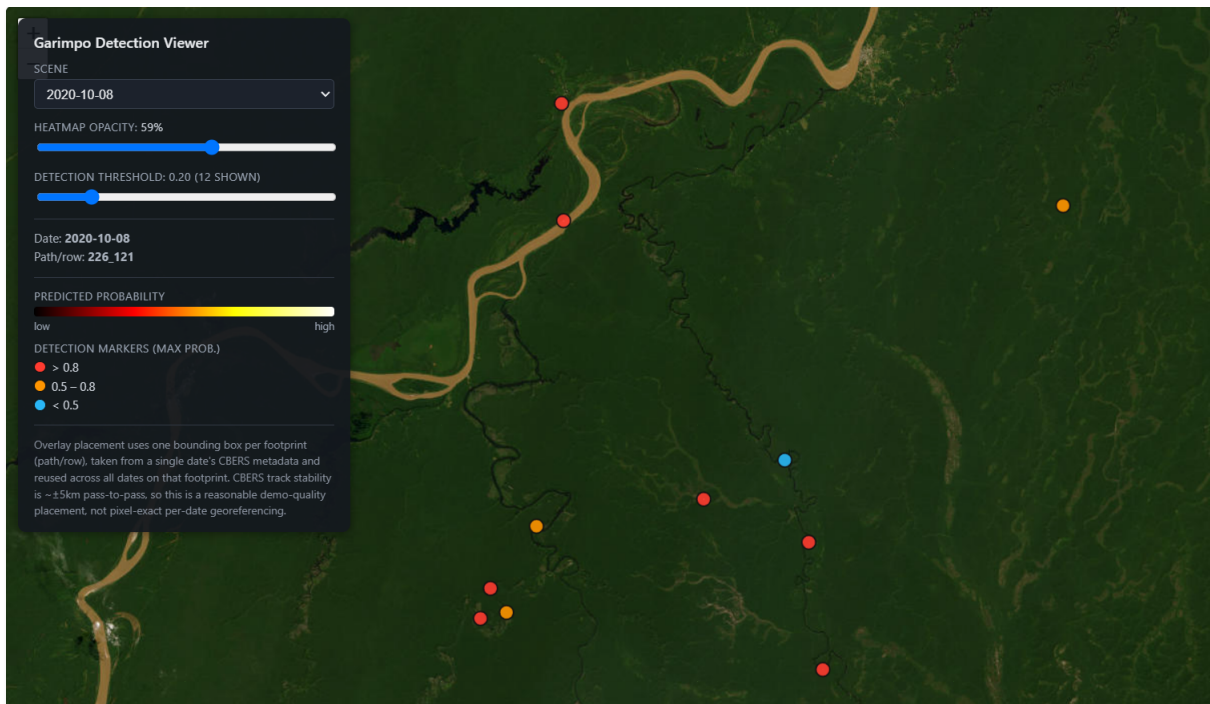


Figure 9: Prototype map interface, showing the confidence-coded detection markers and threshold slider on a reference-date scene. Markers that appear visually distant from the main river channel at this zoom level were investigated (Section 8.4) and found, on closer inspection, to sit near small tributary streams not obvious at this scale — consistent with the water-adjacency finding discussed there, not an inconsistency with it.

10 Next Steps

1. Recover precise, per-date georeferencing (rather than the current one-date-per-footprint approximation) using the recovered raw band GeoTIFFs, and reintroduce the near-infrared band into the training pipeline.
2. Investigate the false-positive outlier at scene 20220620_226_120 (19 of 22 detections false, well above any other scene).

3. Reduce the model’s residual sensitivity to nodata/mosaic-seam artifacts, either by masking such regions at inference time or by targeted hard-negative training on artifact-adjacent tiles.
4. Investigate whether the near-infrared band (recoverable from the raw band GeoTIFFs, per the item above) can help the model separate genuine turbid-water/mining-plume signatures from the spectrally similar natural sediment identified as the largest single explained cause of false positives (Table 6), since RGB true-color imagery alone does not appear sufficient to reliably distinguish them. A river-proximity post-filter was considered and explicitly ruled out as a fix: true positives sit just as consistently near water as false positives do (both approach 100% within a loose buffer), so proximity alone cannot discriminate between them — the spectral similarity itself, not mere location, is the mechanism to address.
5. Investigate the 12 false positives that remain unexplained after the nodata/seam, spectral, and cloud-shadow checks (Section 8.4) — most likely by extending the water/sediment color mask to better cover narrow, shadowed tributary streams, which the current mask under-detects.
6. Build a coordinate-based, on-demand interface: given a user-selected location and time on the map, run inference directly on that location rather than requiring a full pre-computed scene-level heatmap.
7. Address the observed train/validation performance gap (final-epoch train Dice 0.87 vs. validation Dice 0.52), through stronger regularization, more aggressive augmentation, or additional labeled data.
8. Given the two-footprint geographic limitation identified in Section 8, evaluate whether a cross-validation scheme (rotating which scenes are held out) would give a more statistically robust performance estimate than the current single fixed split.
9. Extend the false-positive analysis with a precision–recall curve across multiple probability thresholds, to characterize the precision/recall trade-off beyond the single operating point reported here.
10. Compare model detections against the manually-digitized locations reported by Monteiro (2024), where acquisition dates overlap, as an external validation reference.

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